BOUNDEDNESS CRITERION ON PROJECTIVE SPACE VIA THE BEILINSON FAN

THE STACKS PROJECT AUTHORS

Allowed standard prerequisites (editorial).

- Perfect duality. cohomology-lemma-dual-perfect-complex (cross-chapter, named). (2)-(3) equivalence and the Ext reidentification.
- Endomorphism ring equivalence. perfect-lemma-Pn-module-category (cross-chapter, named). The Beilinson fan P gives an equivalence D(R) -> D_QCoh(O_{P^n}).
- Endomorphism ring compactness. lemma-perfect-for-R L26459 (carded this batch). Condition (2) of that lemma matches the displayed Ext computation.

Source reference. Original reference 08DQ: focused-seeds/geometry/S9-eqv-coherent-on-projective-space/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0CS6: focused-seeds/geometry/S9-eqv-coherent-on-projective-space/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0FYA: sources/stacks-equiv/pinned-original-source.tex, source lines 676–676. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0BQU: focused-seeds/geometry/S9-eqv-coherent-on-projective-space/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08E2: focused-seeds/geometry/S9-eqv-coherent-on-projective-space/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0FDC: focused-seeds/geometry/S9-eqv-coherent-on-projective-space/ref-tags.txt, source lines 9–9. Navigation pointer only: the

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counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 09M1: focused-seeds/geometry/S9-eqv-coherent-on-projective-space/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let k be a field. Let $n \geq 0$. Let $K \in D_{QCoh}(\mathcal{O}_{\mathbf{P}^n_k})$. The following are equivalent*

(1) *K is in $D^b_{Coh}(\mathcal{O}_{\mathbf{P}^n_k})$,*
(2) *$\sum_{i \in \mathbf{Z}} \dim_k H^i(\mathbf{P}^n_k, E \otimes^{\mathbf{L}} K) < \infty$ for each perfect object E of $D(\mathcal{O}_{\mathbf{P}^n_k})$,*
(3) *$\sum_{i \in \mathbf{Z}} \dim_k \mathrm{Ext}^i_{\mathbf{P}^n_k}(E, K) < \infty$ for each perfect object E of $D(\mathcal{O}_{\mathbf{P}^n_k})$,*
(4) *$\sum_{i \in \mathbf{Z}} \dim_k H^i(\mathbf{P}^n_k, K \otimes^{\mathbf{L}} \mathcal{O}_{\mathbf{P}^n_k}(d)) < \infty$ for $d = 0, 1, \ldots, n$.*

Proof. Parts (2) and (3) are equivalent by Cohomology, Lemma 08DQ. If (1) is true, then for E perfect the derived tensor product $E \otimes^{\mathbf{L}} K$ is in $D^b_{Coh}(\mathcal{O}_{\mathbf{P}^n_k})$ and we see that (2) holds by Derived Categories of Schemes, Lemma 08E2. It is clear that (2) implies (4) as $\mathcal{O}_{\mathbf{P}^n_k}(d)$ can be viewed as a perfect object of the derived category of $\mathbf{P}^n_k$. Thus it suffices to prove that (4) implies (1).

Assume (4). Let R be as in Lemma 0FYA. Let $P = \bigoplus_{d=0,\ldots,n} \mathcal{O}_{\mathbf{P}^n_k}(-d)$. Recall that $R = \mathrm{End}_{\mathbf{P}^n_k}(P)$ whereas all other self-Exts of P are zero and that P determines an equivalence $- \otimes^{\mathbf{L}} P : D(R) \to D_{QCoh}(\mathcal{O}_{\mathbf{P}^n_k})$ by Derived Categories of Schemes, Lemma 0BQU. Say K corresponds to L in $D(R)$. Then

$$\begin{aligned} H^i(L) &= \mathrm{Ext}^i_{D(R)}(R, L) \\ &= \mathrm{Ext}^i_{\mathbf{P}^n_k}(P, K) \\ &= H^i(\mathbf{P}^n_k, K \otimes P^\vee) \\ &= \bigoplus\nolimits_{d=0,\ldots,n} H^i(\mathbf{P}^n_k, K \otimes \mathcal{O}(d)) \end{aligned}$$

by Differential Graded Algebra, Lemma 0CS6 (and the fact that $- \otimes^{\mathbf{L}} P$ is an equivalence) and Cohomology, Lemma 08DQ. Thus our assumption (4) implies that L satisfies condition (2) of Lemma 0FYA and hence is a compact object of $D(R)$. Therefore K is a compact object of $D_{QCoh}(\mathcal{O}_{\mathbf{P}^n_k})$. Thus K is perfect by Derived Categories of Schemes, Proposition 09M1. Since $D_{perf}(\mathcal{O}_{\mathbf{P}^n_k}) = D^b_{Coh}(\mathcal{O}_{\mathbf{P}^n_k})$ by Derived Categories of Schemes, Lemma 0FDC we conclude (1) holds. □